

Total No. of Questions : 6]

SEAT No. :

P3661

[Total No. of Pages : 2

APR - 15 / Engg. - 146

T.E. INFORMATION TECHNOLOGY (In Sem - Semester - II)

Design & Analysis of Algorithms

(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) Define best-case, worst-case and average-case efficiency. Is average-case efficiency is an average of best-case and worst-case efficiencies? **[5]**

b) State tiling problem and prove it by mathematical induction. **[5]**

OR

Q2) a) With the help of example, explain general strategy for analyzing time efficiency of non-recursive algorithm. **[5]**

b) Solve following recurrence relation using backward substitution method. **[5]**

$$T(n) = T(n-1) + 1$$

$$T(0) = 0$$

Q3) a) Explain the concept of divide and conquer technique. Write Master theorem. **[5]**

b) Give recurrence relation for binary search and solve the same. **[5]**

OR

P.T.O.

Q4) a) What is Huffman code? [2]

b) Construct Huffman tree for following data and obtain its Huffman code.[8]

Character	A	B	C	D	E	F
Frequency	0.5	0.35	0.5	0.1	0.4	0.2

Q5) a) What is travelling salesperson problem? How it can be solved by dynamic programming strategy? [5]

b) Apply the bottom-up dynamic programming algorithm to the following instance of the knapsack problem where capacity of knapsack i.e. $W = 5$ [5]

Item	Weight	Value
1	2	12
2	1	10
3	3	20
4	2	15

OR

Q6) a) What is optimal binary search tree? [2]

b) Let $N=3$, and $\{a_1, a_2, a_3\} = \{\text{do, if, while}\}$. Let $p(1:3) = \{0.5, 0.1, 0.05\}$ and $q(0:3) = \{0.15, 0.1, 0.05, 0.05\}$. Compute and construct OBST for above value using Dynamic Programming. [8]
